

INDUSTRY PROBLEM SOLVING TASK

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Q1. Autonomous Vehicle Vision System Selection

Constraints

- Accuracy must be $\geq 90\%$ (below this, the system fails)
- Latency must be ≤ 70 ms (above this, it is unacceptable)
- Hardware supports only moderate computational resources

Given Data

Method	Accuracy	Latency	Computation
Viola-Jones	82%	30 ms	Low
YOLO	93%	60 ms	Moderate
Deep Learning	97%	90 ms	High

Constraint-wise Comparison

- Viola-Jones: Accuracy = $82\% < 90\%$ → fails the accuracy constraint, despite excellent latency (30 ms) and low computation. Eliminated.
- YOLO: Accuracy = $93\% \geq 90\%$ (pass); Latency = $60 \text{ ms} \leq 70 \text{ ms}$ (pass); Computation = Moderate, matching available hardware (pass). Satisfies all three constraints.
- Deep Learning: Accuracy = $97\% \geq 90\%$ (pass); Latency = $90 \text{ ms} > 70 \text{ ms}$ → fails the latency constraint; Computation = High, exceeding moderate hardware capacity (fails). Eliminated.

Evaluation & Decision

Only YOLO satisfies all three constraints simultaneously — it clears the minimum accuracy threshold, stays within the latency budget, and matches the vehicle's moderate computational resources. Viola-Jones fails on accuracy alone, and Deep Learning fails on both latency and computation despite having the highest accuracy.

Justification

In an autonomous vehicle context, all three constraints are hard requirements, not preferences — a system that is too slow (Deep Learning) or too inaccurate (Viola-Jones) is unsafe regardless of its strengths elsewhere. YOLO is the only method that meets the accuracy bar while remaining fast enough for real-time driving decisions and compatible with the vehicle's onboard hardware, making it the correct and only valid selection.

Q2. Smart Factory Product Inspection Selection

Constraints

- Accuracy must be $\geq 95\%$ (below this, defects may be missed)
- Processing time must be ≤ 50 ms (above this, production is affected)
- Hardware supports only moderate computation

Given Data

Method	Accuracy	Processing Time	Computation
Traditional Image Analysis	91%	25 ms	Low
YOLO	96%	45 ms	Moderate
Deep Learning	98%	75 ms	High

Constraint-wise Comparison

- Traditional Image Analysis: Accuracy = $91\% < 95\%$ → fails the accuracy constraint, even though processing time (25 ms) and computation (low) are well within limits. Eliminated.
- YOLO: Accuracy = $96\% \geq 95\%$ (pass); Processing time = $45\text{ ms} \leq 50\text{ ms}$ (pass); Computation = Moderate, matching available hardware (pass). Satisfies all three constraints.
- Deep Learning: Accuracy = $98\% \geq 95\%$ (pass); Processing time = $75\text{ ms} > 50\text{ ms}$ → fails the processing-time constraint; Computation = High, exceeding moderate hardware capacity (fails). Eliminated.

Evaluation & Decision

As in the first scenario, YOLO is the only method that satisfies all three constraints together. Traditional Image Analysis is disqualified by insufficient accuracy, which risks letting defective products pass inspection undetected. Deep Learning offers the highest accuracy (98%) but its processing time and computational demand exceed what the production line and hardware can support.

Justification

A production-line inspection system must not slow down manufacturing throughput, so processing time is a hard ceiling, not a target to exceed. YOLO meets the required 95% accuracy threshold to reliably catch defects while staying under the 50 ms processing budget and operating within the factory's moderate-computation hardware — making it the only method that is simultaneously accurate enough, fast enough, and resource-compatible for real-time deployment on the production line.